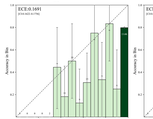

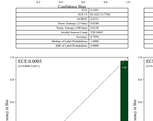

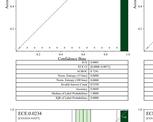

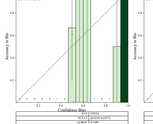

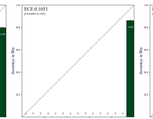

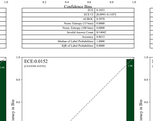

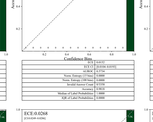

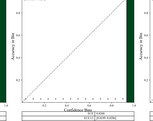

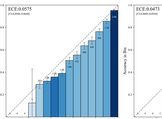

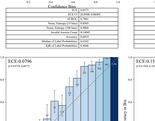

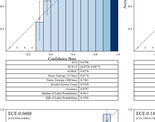

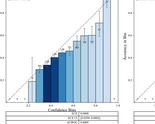

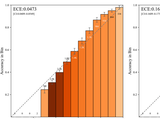

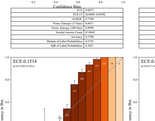

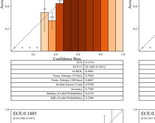

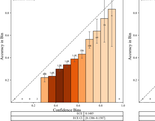

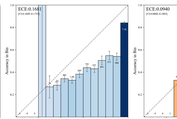

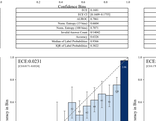

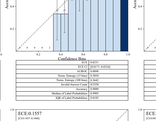

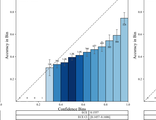

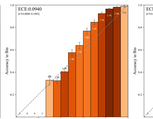

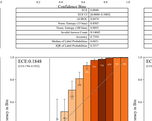

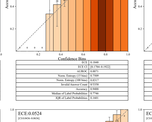

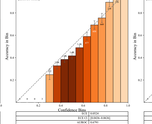

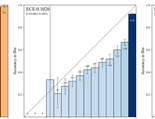

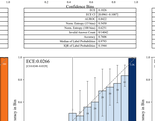

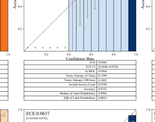

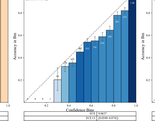

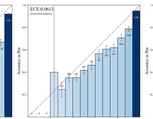

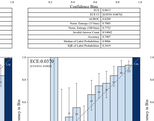

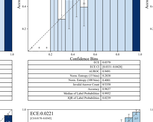

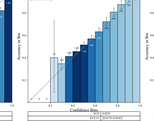

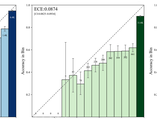

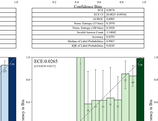

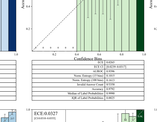

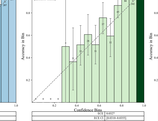

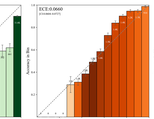

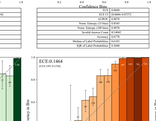

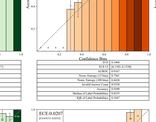

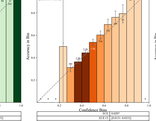

Calibration of Normalized Label Probabilities for Prompt 1 (Most Probable Label per Question Only)

MMLU

	ECE	ECE CI	AUROC	Norm. Entropy (15 bins)	Norm. Entropy (100 bins)	Invalid Answer Count	Accuracy	Median of Label Probabilities	IQR of Label Probabilities
gpt-oss-20b	0.1691	[0.1622–0.1756]	0.6231	0.0180	0.0138	328/14042	0.7956	1.0000	0.0000
gpt-oss-120b	0.1053	[0.0993–0.1107]	0.5970	0.0000	0.0000	0/14042	0.8613	1.0000	0.0000
Ministral-8B	0.0575	[0.0508–0.0649]	0.7861	0.8505	0.9068	0/14042	0.6014	0.6342	0.4446
Mistral-Nemo-Base	0.0473	[0.0409–0.0549]	0.7740	0.8411	0.8999	0/14042	0.5708	0.5192	0.3021
Mistral-Nemo-7B	0.1681	[0.1609–0.1755]	0.7861	0.6604	0.7071	0/14042	0.6326	0.9366	0.3822
Mistral-Small-3.1-24B	0.0940	[0.0880–0.1003]	0.8474	0.8503	0.9055	0/14042	0.7354	0.6831	0.3517
Mistral-Small-3.2-24B	0.1026	[0.0961–0.1087]	0.8422	0.5450	0.6251	0/14042	0.7606	0.9793	0.1944
Magistral-Small	0.0613	[0.0554–0.0676]	0.8289	0.7005	0.7752	0/14042	0.7407	0.9066	0.3419
Magistral-Small-24B-Reasoning	0.0874	[0.0825–0.0934]	0.8003	0.2970	0.3928	1/14042	0.8393	0.9967	0.0245
Llama-3.1-70B	0.0660	[0.0604–0.0727]	0.8078	0.8545	0.9070	0/14042	0.6778	0.6183	0.3040
Llama-3.3-70B	0.1548	[0.1483–0.1615]	0.8294	0.1206	0.1308	0/14042	0.7997	1.0000	0.0000
DeepSeek-Llama-70B	0.0752	[0.0692–0.0819]	0.6092	0.6636	0.7975	77/14042	0.8222	0.9047	0.1931
Llama-4-Scout-17B-16E	0.1093	[0.1028–0.1158]	0.8360	0.5887	0.6563	0/14042	0.7394	0.9686	0.2502
Llama-4-Scout-17B	0.1749	[0.1685–0.1816]	0.8095	0.0601	0.0665	0/14042	0.7864	1.0000	0.0000
Qwen3-30B-A3B-Base	0.0408	[0.0351–0.0472]	0.8233	0.8239	0.8870	0/14042	0.7450	0.7917	0.3201
Qwen3-30B-A3B-Instruct	0.1469	[0.1409–0.1540]	0.8500	0.2587	0.2914	0/14042	0.7869	1.0000	0.0067
Qwen3-30B-A3B-Thinking	0.0313	[0.0249–0.0384]	0.7273	0.8540	0.9115	5/14042	0.6961	0.6810	0.3480
DeepSeek-Qwen-32B	0.0462	[0.0420–0.0528]	0.6851	0.6028	0.7531	86/14042	0.8249	0.9171	0.1233
gemma-3-27b-pt	0.0227	[0.0174–0.0302]	0.7677	0.8556	0.9045	0/14042	0.6824	0.6564	0.3528
google/gemma-3-27b	0.2081	[0.2007–0.2149]	0.8003	0.0975	0.1100	0/14042	0.7491	1.0000	0.0000

ArcReasoning

	ECE	ECE CI	AUROC	Norm. Entropy (15 bins)	Norm. Entropy (100 bins)	Invalid Answer Count	Accuracy	Median of Label Probabilities	IQR of Label Probabilities
gpt-oss-20b	0.0003	[0.0000–0.0071]	0.7241	0.0000	0.0000	0/3358	0.9669	1.0000	0.0000
gpt-oss-120b	0.0152	[0.0104–0.0193]	0.5734	0.0000	0.0000	0/3358	0.9818	1.0000	0.0000
Ministral-8B	0.0796	[0.0728–0.0877]	0.9278	0.6174	0.7581	0/3358	0.9297	0.9411	0.1956
Mistral-Nemo-Base	0.1514	[0.1420–0.1651]	0.8401	0.7864	0.8687	0/3358	0.7564	0.6110	0.2304
Mistral-Nemo-7B	0.0231	[0.0171–0.0324]	0.8898	0.3054	0.3642	0/3358	0.9089	0.9985	0.0183
Mistral-Small-3.1-24B	0.1848	[0.1784–0.1922]	0.8871	0.7509	0.8317	0/3358	0.9488	0.7746	0.1601
Mistral-Small-3.2-24B	0.0266	[0.0248–0.0329]	0.9564	0.1299	0.1882	0/3358	0.9732	0.9994	0.0024
Magistral-Small	0.0370	[0.0331–0.0428]	0.9491	0.2838	0.4001	0/3358	0.9637	0.9952	0.0239
Magistral-Small-24B-Reasoning	0.0265	[0.0239–0.0317]	0.9396	0.1015	0.1613	0/3358	0.9792	0.9990	0.0025
Llama-3.1-70B	0.1464	[0.1392–0.1538]	0.9167	0.7561	0.8420	0/3358	0.9390	0.8335	0.1847
Llama-3.3-70B	0.0068	[0.0029–0.0126]	0.9380	0.0253	0.0299	0/3358	0.9663	1.0000	0.0000
DeepSeek-Llama-70B	0.0697	[0.0616–0.0787]	0.8109	0.6340	0.7778	2/3358	0.9249	0.9135	0.1640
Llama-4-Scout-17B-16E	0.0291	[0.0256–0.0375]	0.9318	0.2676	0.3800	0/3358	0.9407	0.9957	0.0207
Llama-4-Scout-17B	0.0052	[0.0021–0.0121]	0.8892	0.0172	0.0153	0/3358	0.9604	1.0000	0.0000
Qwen3-30B-A3B-Base	0.0799	[0.0749–0.0859]	0.9265	0.5708	0.7306	0/3358	0.9601	0.9207	0.1005
Qwen3-30B-A3B-Instruct	0.0206	[0.0158–0.0257]	0.9145	0.0493	0.0470	0/3358	0.9732	1.0000	0.0000
Qwen3-30B-A3B-Thinking	0.1239	[0.1119–0.1369]	0.8717	0.8319	0.8960	0/3358	0.6334	0.6919	0.4095
DeepSeek-Qwen-32B	0.0514	[0.0477–0.0576]	0.8149	0.4062	0.6324	8/3358	0.9722	0.9548	0.0663
gemma-3-27b-pt	0.1447	[0.1365–0.1531]	0.8760	0.7614	0.8287	0/3358	0.9276	0.7959	0.2736
google/gemma-3-27b	0.0035	[0.0014–0.0103]	0.8432	0.0170	0.0161	0/3358	0.9628	1.0000	0.0000

GSM8KMC

	ECE	ECE CI	AUROC	Norm. Entropy (15 bins)	Norm. Entropy (100 bins)	Invalid Answer Count	Accuracy	Median of Label Probabilities	IQR of Label Probabilities
gpt-oss-20b	0.0234	[0.0210–0.0257]	0.7309	0.0058	0.0045	16/7468	0.9887	1.0000	0.0000
gpt-oss-120b	0.0268	[0.0249–0.0286]	0.5480	0.0000	0.0000	0/7468	0.9934	1.0000	0.0000
Ministral-8B	0.0488	[0.0394–0.0605]	0.6005	0.6943	0.8116	0/7468	0.4059	0.4310	0.1459
Mistral-Nemo-Base	0.1485	[0.1386–0.1587]	0.6010	0.6652	0.7961	0/7468	0.3243	0.4619	0.1412
Mistral-Nemo-7B	0.1557	[0.1457–0.1686]	0.6010	0.8305	0.8977	0/7468	0.4298	0.5510	0.2469
Mistral-Small-3.1-24B	0.0524	[0.0436–0.0636]	0.6791	0.7828	0.8684	0/7468	0.4889	0.5163	0.2043
Mistral-Small-3.2-24B	0.0637	[0.0540–0.0742]	0.7439	0.8486	0.9074	0/7468	0.6248	0.6803	0.3447
Magistral-Small	0.0221	[0.0170–0.0345]	0.6859	0.8250	0.8932	0/7468	0.5787	0.5203	0.2349
Magistral-Small-24B-Reasoning	0.0327	[0.0310–0.0355]	0.9238	0.1454	0.2916	0/7468	0.9834	0.9951	0.0108
Llama-3.1-70B	0.0207	[0.0151–0.0333]	0.6418	0.6596	0.7912	0/7468	0.4716	0.4317	0.1308
Llama-3.3-70B	0.3001	[0.2889–0.3103]	0.7220	0.3027	0.3442	0/7468	0.6260	0.9998	0.0206
DeepSeek-Llama-70B	0.0760	[0.0673–0.0840]	0.8088	0.8044	0.8798	48/7468	0.7992	0.7260	0.3028
Llama-4-Scout-17B-16E	0.1553	[0.1441–0.1671]	0.6704	0.8439	0.9060	0/7468	0.5408	0.6897	0.3199
Llama-4-Scout-17B	0.3711	[0.3595–0.3830]	0.6603	0.1729	0.1834	0/7468	0.5766	1.0000	0.0004
Qwen3-30B-A3B-Base	0.0418	[0.0336–0.0522]	0.7303	0.8461	0.9039	0/7468	0.6742	0.6236	0.2888
Qwen3-30B-A3B-Instruct	0.1829	[0.1732–0.1926]	0.7733	0.4341	0.4925	0/7468	0.7171	0.9948	0.1012
Qwen3-30B-A3B-Thinking	0.0363	[0.0346–0.0380]	0.6802	0.1341	0.4170	3/7468	0.9945	0.9851	0.0232
DeepSeek-Qwen-32B	0.2025	[0.1934–0.2119]	0.8091	0.8201	0.8875	91/7468	0.5196	0.6886	0.2811
gemma-3-27b-pt	0.1408	[0.1304–0.1515]	0.7156	0.8105	0.8831	0/7468	0.5083	0.6277	0.2338
google/gemma-3-27b	0.2908	[0.2813–0.3013]	0.7286	0.1816	0.2043	0/7468	0.6559	1.0000	0.0009

GPQA

	ECE	ECE CI	AUROC	Norm. Entropy (15 bins)	Norm. Entropy (100 bins)	Invalid Answer Count	Accuracy	Median of Label Probabilities	IQR of Label Probabilities
gpt-oss-20b	0.3092	[0.2664–0.3571]	0.6355	0.0000	0.0000	51/448	0.6574	1.0000	0.0000
gpt-oss-120b	0.3037	[0.2613–0.3506]	0.5711	0.0000	0.0034	0/448	0.6629	1.0000	0.0000
Ministral-8B	0.1211	[0.0876–0.1643]	0.5908	0.6705	0.7773	0/448	0.2946	0.3708	0.1392
Mistral-Nemo-Base	0.1976	[0.1580–0.2335]	0.5859	0.6384	0.7666	0/448	0.2567	0.4331	0.1210
Mistral-Nemo-7B	0.2304	[0.1896–0.2753]	0.5964	0.8471	0.8920	0/448	0.3438	0.5120	0.3139
Mistral-Small-3.1-24B	0.1225	[0.0884–0.1661]	0.6504	0.6714	0.7897	0/448	0.3460	0.4332	0.1306
Mistral-Small-3.2-24B	0.2278	[0.1882–0.2729]	0.7032	0.8644	0.9010	0/448	0.3951	0.5734	0.3589
Magistral-Small	0.1277	[0.0945–0.1735]	0.6149	0.8275	0.8788	0/448	0.4152	0.4815	0.2618
Magistral-Small-24B-Reasoning	0.4031	[0.3585–0.4614]	0.5658	0.5988	0.6165	122/448	0.4356	0.9790	0.2817
Llama-3.1-70B	0.1789	[0.1416–0.2260]	0.5697	0.6473	0.7694	0/448	0.2946	0.4605	0.1245
Llama-3.3-70B	0.4650	[0.4214–0.5107]	0.7217	0.2940	0.3046	0/448	0.4643	1.0000	0.0141
DeepSeek-Llama-70B	0.1437	[0.1134–0.1920]	0.6679	0.7511	0.8379	21/448	0.6674	0.8654	0.2371
Llama-4-Scout-17B-16E	0.2092	[0.1723–0.2537]	0.6489	0.8484	0.8898	0/448	0.3728	0.5318	0.2618
Llama-4-Scout-17B	0.5342	[0.4893–0.5771]	0.6723	0.1830	0.2076	0/448	0.4085	1.0000	0.0009
Qwen3-30B-A3B-Base	0.1659	[0.1229–0.2112]	0.5847	0.7579	0.8330	0/448	0.3616	0.4837	0.1862
Qwen3-30B-A3B-Instruct	0.3673	[0.3250–0.4158]	0.6789	0.6999	0.7521	0/448	0.4286	0.8522	0.3401
Qwen3-30B-A3B-Thinking	0.1693	[0.1368–0.2152]	0.5945	0.6349	0.7683	50/448	0.7136	0.9171	0.1694
DeepSeek-Qwen-32B	0.2268	[0.1866–0.2737]	0.6283	0.7921	0.8593	58/448	0.5487	0.8233	0.3295
gemma-3-27b-pt	0.2643	[0.2250–0.3084]	0.5645	0.6633	0.7788	0/448	0.2991	0.5414	0.1250
google/gemma-3-27b	0.5799	[0.5370–0.6225]	0.6493	0.2988	0.3288	0/448	0.3482	0.9998	0.0194